

Variance and Covariance

PHS Launch!

Project Hail Mary running example

PHS 2000A Orientation

Goals:

By the end of this session, you should be familiar with:

- Variance as a measure of spread around a mean
- Why variance uses squared deviations
- The difference between population and sample variance / standard deviation
- Covariance as a measure of how two variables vary together
- The difference between population and sample covariance
- Why variance and covariance matter for methods you will see later in PHS

Key terms and definitions will be highlighted **like this!**

Running example: Astrophage in the lab

Scientists on Earth have just discovered Astrophage. To study how Astrophage responds to light exposure, Grace's lab divided one sample of Astrophage into five petri dishes. Each dish was exposed to the same standardized light source for 30 seconds. After exposure, Grace measured how much additional energy each sample stored and how much infrared radiation it emitted. He repeated this experiment three times, using three different samples of Astrophage.

We track three variables:

- **Light Exposure (X):** duration of exposure to the energy source, in seconds (s); held constant at 30 seconds
- **Energy Stored (Y):** additional energy stored after exposure, in Joules (J)
- **Infrared Radiation Emitted (Z):** infrared radiation emitted after exposure, in watts per square meter (W/m^2)

Experiment A			
Dish	X: Light Exposure (s)	Y: Energy Stored (J)	Z: Infrared Radiation Emitted (W/m^2)
1	30	20	12
2	30	40	32
3	30	50	45
4	30	70	62
5	30	70	64

Experiment B			
Dish	X: Light Exposure (s)	Y: Energy Stored (J)	Z: Infrared Radiation Emitted (W/m^2)
1	30	46	40
2	30	48	42
3	30	50	43
4	30	52	44
5	30	54	46

Experiment C			
Dish	X: Light Exposure (s)	Y: Energy Stored (J)	Z: Infrared Radiation Emitted (W/m^2)
1	30	20	13
2	30	35	28
3	30	50	43
4	30	65	58
5	30	80	73

Start with the center: the mean

The **sample mean** tells us where the average of the observed values in the sample.

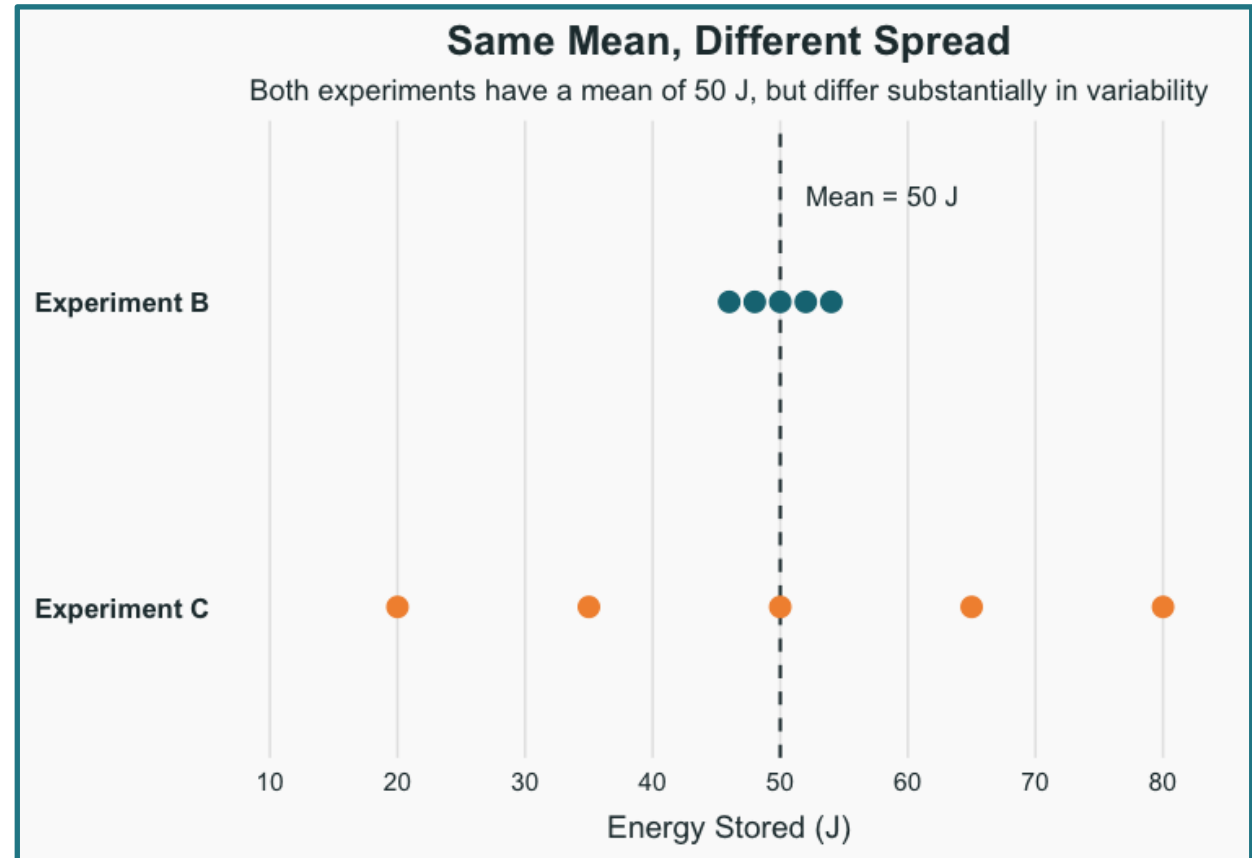
Experiment B and C results:

$$\bar{y}_B = \frac{46 + 48 + 50 + 52 + 54}{5} = 50 \text{ J}$$

$$\bar{y}_C = \frac{20 + 35 + 50 + 65 + 80}{5} = 50 \text{ J}$$

Key idea:

The mean describes the center, but it does not tell us how spread out the observations are.



How far is each observation from the sample mean?

A **deviation** compares each observation with the sample mean.

Experiment B: low spread

Dish	Energy Stored (J) $y_{i,B}$	Deviation (J) $y_{i,B} - \bar{y}_B$
1	46	-4
2	48	-2
3	50	0
4	52	+2
5	54	+4

Experiment C: high spread

Dish	Energy Stored (J) $y_{i,C}$	Deviation (J) $y_{i,C} - \bar{y}_C$
1	20	-30
2	35	-15
3	50	0
4	65	+15
5	80	+30

Interpretation:

Experiment B has small deviations from 50 J; Experiment C has much larger deviations from the same mean.

Why not just average the deviations?

Because positive and negative deviations around the sample mean cancel:

Experiment B Deviation:

$$\frac{-4 + -2 + 0 + 2 + 4}{5} = 0J$$

Experiment C Deviation:

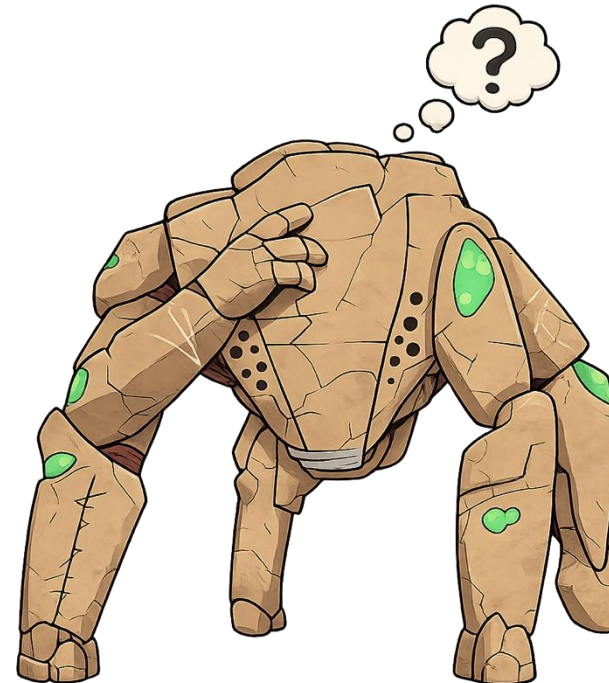
$$\frac{-30 + -15 + 0 + 15 + 30}{5} = 0J$$

A dataset can be very spread out and still have a mean deviation equal to zero.

Solution:

Square the deviations before averaging them.

Squaring prevents positive and negative deviations from canceling and makes large deviations count more.



Variance: population and sample versions

The **variance** is a measure of how much observations vary, or spread out, around their mean.

Population variance

Describes the spread of the entire population.

$$\sigma_X^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)^2$$

Sample variance

Uses observed data to estimate the population variance.

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Same concept in both cases: subtract the appropriate mean → square → sum → divide

Why n – 1?

Because the sample mean is estimated from the same data, one degree of freedom is used. Dividing by n – 1 corrects the sample variance when estimating population variance.

Worked example: same mean, different sample variance

Both experiments have the same mean energy stored: $\bar{y} = 50$ J.

Experiment B: low spread

Dish	Deviation (J) $y_{i,B} - \bar{y}_B$	Squared deviation (J ²) $(y_{i,B} - \bar{y}_B)^2$
1	-4	16
2	-2	4
3	0	0
4	+2	4
5	+4	16
Sum	—	40

Sample variance

$$\text{Experiment B: } s_B^2 = \frac{40}{5-1} = 10 \text{ J}^2$$

Experiment C: high spread

Dish	Deviation (J) $y_{i,C} - \bar{y}_C$	Squared deviation (J ²) $(y_{i,C} - \bar{y}_C)^2$
1	-30	900
2	-15	225
3	0	0
4	+15	225
5	+30	900
Sum	—	2,250

$$\text{Experiment C: } s_C^2 = \frac{2250}{5-1} = 562.5 \text{ J}^2$$

Same mean, but very different spread: Experiment C has a much larger sample variance than Experiment B.

Standard deviation: population and sample versions

The **standard deviation** is the square root of the variance.

Population standard deviation

Describes spread in the entire population.

$$\sigma_Y = \sqrt{\sigma_Y^2}$$

Sample standard deviation

Uses observed data to estimate the population standard deviation.

$$s_Y = \sqrt{s_Y^2}$$

Same concept in both cases: take the square root of the corresponding variance.

Worked example: sample standard deviation

Experiment B

$$s_B = \sqrt{10} \approx 3.16 \text{ J}$$

Experiment C

$$s_C = \sqrt{562.5} \approx 23.72 \text{ J}$$

Why is this useful?

Variance is expressed in squared units (J^2). Standard deviation returns variability to the original units (Joules).

Interpretation

Experiment C has a much larger standard deviation than Experiment B, so its energy-storage values are much more spread out around the sample mean.

From variance to covariance

Variance

How does **ONE** variable vary around its own mean?

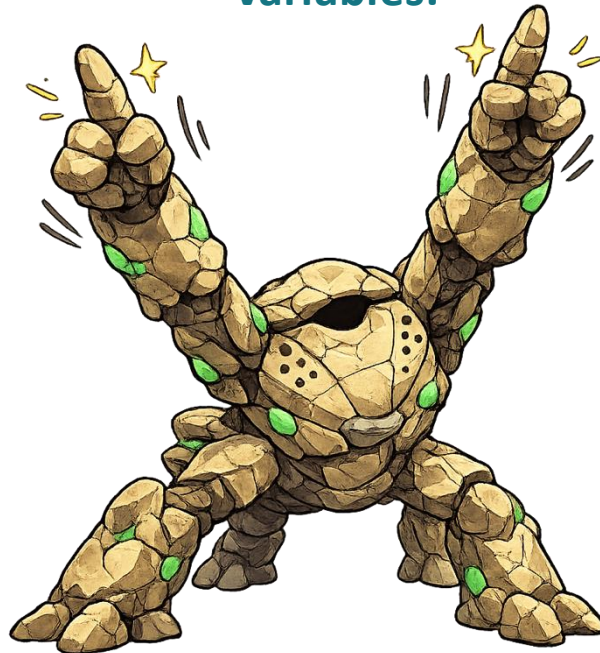
$$\text{Var}(X) = E[(X - E[X])(X - E[X])]$$

Covariance

How do **TWO** variables vary around their means together?

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Covariance has the same basic logic as variance — but the two deviations come from different variables.



Covariance also has population and sample versions

Population covariance

Describes how X and Y vary together in the population.

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y)$$

Sample covariance

Uses paired sample observations to estimate population covariance.

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

The conceptual operation is unchanged; the notation tells us whether we are describing a population or estimating it from a sample.

Reintroducing Experiment A: two measured responses

So far, we used energy stored (Y) to understand spread. Now we bring back the second response measured in Experiment A: infrared radiation emitted (Z).

Covariance question

When Energy Stored is above or below its mean, does Infrared Radiation Emitted tend to be above or below its mean in the same direction?

Variables in focus

Y: Energy Stored, in Joules (J)

Z: Infrared Radiation Emitted, in watts per square meter (W/m^2)

Experiment A			
Dish	X: Light Exposure (s)	Y: Energy Stored (J)	Z: Infrared Radiation Emitted (W/m^2)
1	30	20	12
2	30	40	32
3	30	50	45
4	30	70	62
5	30	70	64

Worked example: sample covariance

Dish	Y: Energy Stored (J)	Deviation (J) $y_i - \bar{y}$	Z: Infrared Radiation Emitted (W/m ²)	Infrared Radiation Emitted Deviation (W/m ²) $z_i - \bar{z}$	Product of Deviations (J·W/m ²) $(y_i - \bar{y})(z_i - \bar{z})$
1	20	-30	12	-31	930
2	40	-10	32	-11	110
3	50	0	45	+2	0
4	70	+20	62	+19	380
5	70	+20	64	+21	420
Sum	—	—	—	—	1,840

Means used in this calculation:

$$\bar{y} = 50 \text{ J}$$

$$\bar{z} = 43 \text{ W/m}^2$$

Because these five dishes are a sample:

$$s_{YZ} = \frac{\sum_{i=1}^n (y_i - \bar{y})(z_i - \bar{z})}{n - 1}$$

$$s_{YZ} = \frac{1840}{5 - 1} = 460 \text{ J} \cdot \text{W/m}^2$$

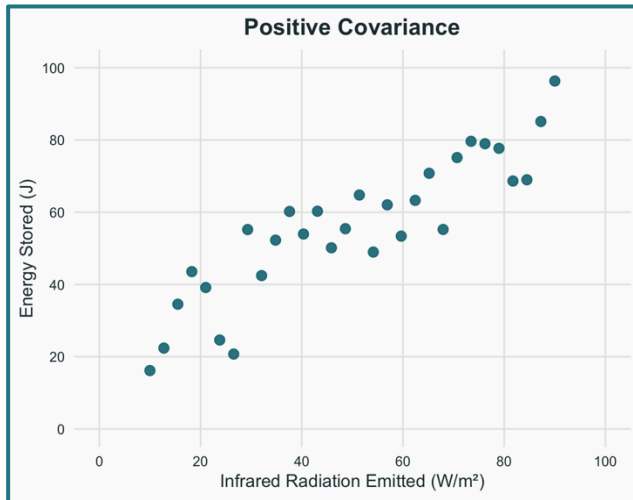
Interpretation:

The sample covariance is positive: above-average energy stored tends to occur with above-average infrared radiation emitted.

How do we interpret covariance?

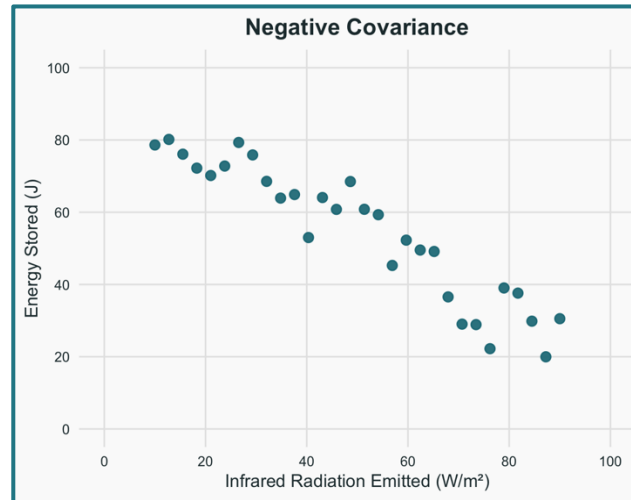
Positive covariance

Higher-than-average Y tends to occur with higher-than-average Z.



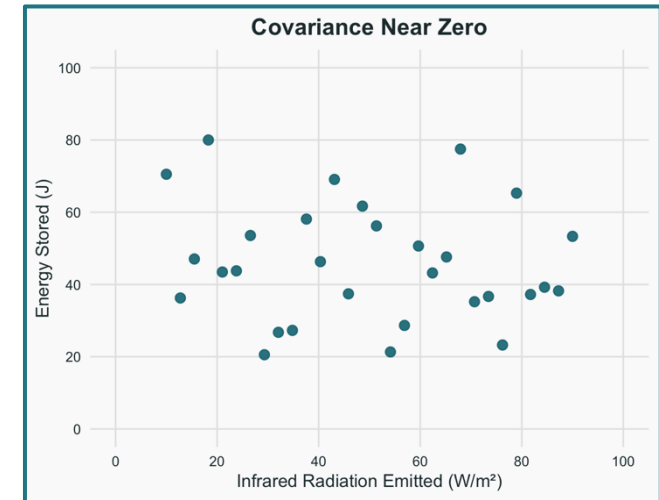
Negative covariance

Higher-than-average Y tends to occur with lower-than-average Z.



Near zero covariance

There is little overall linear co-variation between Y and Z.



Important: covariance near zero does not rule out a nonlinear relationship.

Rules of covariance

Basic identities

Rule	Formula	What it means
Constant	$\text{Cov}(X, a) = 0$	A constant has no variation, so there is no joint variation with X.
Self	$\text{Cov}(X, X) = \text{Var}(X)$	Variance is covariance with itself.
Symmetry	$\text{Cov}(X, Y) = \text{Cov}(Y, X)$	The order of the variables does not matter.

Transformations

Rule	Formula	What it means
Scaling	$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$	Multiplying variables by constants scales the covariance.
Shifting	$\text{Cov}(X + a, Y + b) = \text{Cov}(X, Y)$	Adding constants changes the means, but not the covariance.

Given constants a and b

Mini-example

Suppose $\text{Cov}(X, Y) = 4$

Scaling:

$$\text{Cov}(2X, 3Y) = (2)(3) \text{Cov}(X, Y) = 24$$

Shifting:

$$\text{Cov}(X + 5, Y - 10) = \text{Cov}(X, Y) = 4$$

Key Takeaway

Covariance changes with scale, but not with shifts in location

Why covariance is not enough

Direction

The sign tells us whether X and Y tend to move together or in opposite directions.

Magnitude

The numerical value depends on the units and scale of X and Y, so $460 \text{ J}\cdot\text{W}/\text{m}^2$ is not inherently “large” or “small.”

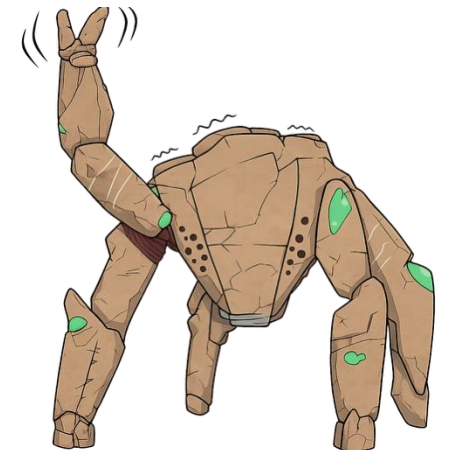
Association, not causation

Covariance describes co-variation in the observed data; by itself, it does not establish a causal relationship.

Why correlation comes next

Correlation standardizes covariance by the standard deviations of X and Y, producing a unitless quantity on a common scale.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$



Covariance gives direction; correlation makes the magnitude interpretable.

Take-home messages

Population vs. sample

Parameters describe populations; sample statistics estimate those parameters.

Variance

Measures how one variable spreads around its mean.

Standard deviation

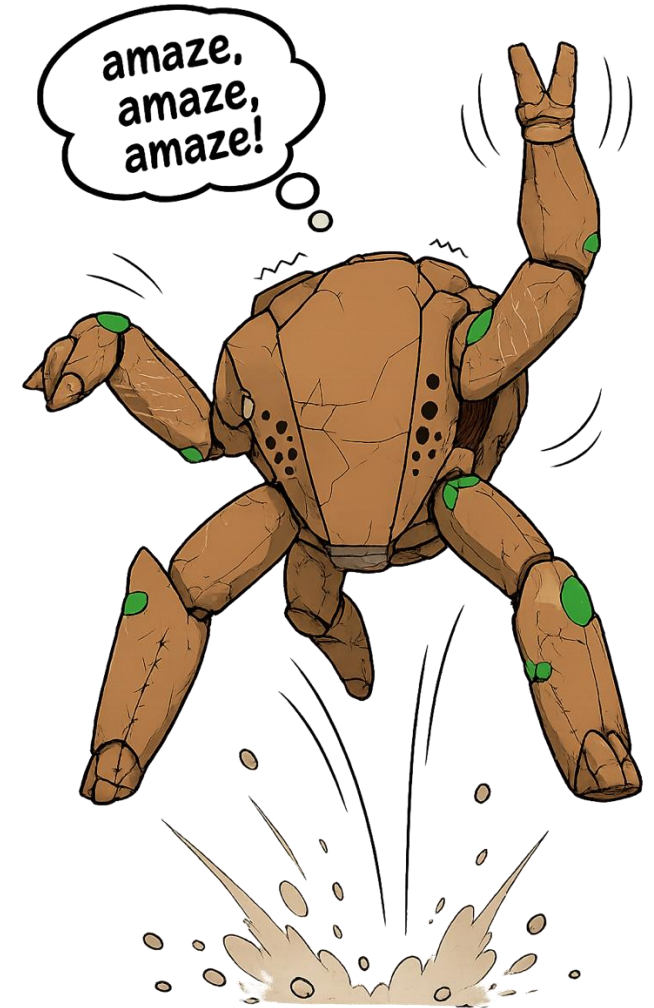
Returns that variability to the variable's original units.

Covariance

Measures whether two variables tend to deviate from their means together.

Why it matters

These ideas underpin uncertainty, regression, correlated data, multilevel models, and measurement.



Variance = variation in one variable. Covariance = variation shared by two variables.